No Single Uncertainty-Quantification Method Reliably Flags Domain-Shift Failure in a Cervical Cytology Classifier

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Abstract. A vision transformer that classifies cervical cytology images with 96%+ in-distribution accuracy but collapses to near- or below-chance accuracy under real cross-institution domain shift – as we previously showed EfficientViT-M0 does on the Mendeley LBC dataset – is only as dangerous as its confidence is falsely high when it fails. We compare four uncertainty-quantification methods on that exact backbone, using each method's own cheapest native confidence signal (temperature-scaled max-softmax, MC Dropout entropy, evidential vacuity, deep-ensemble variance), evaluated in distribution and on both a moderate (Herlev) and severe (Mendeley LBC) external shift. In distribution, three of the four methods calibrate well (expected calibration error 0.012–0.013); evidential deep learning's raw belief masses are markedly less calibrated (0.159) at matched point accuracy. Under moderate shift, a five-member deep ensemble is unambiguously the best method by every metric measured. Under severe shift – the condition that matters most, since it is where the backbone is known to fail – the ranking inverts: MC Dropout's entropy is the only signal that substantially improves shift-failure detection over the uncalibrated baseline (risk-coverage AUC 0.515 to 0.285), while the deep ensemble's advantage does not transfer (0.473, barely better than baseline) despite being the clear winner on the milder shift. No method reaches a reliable abstention operating point on the severe shift condition: even MC Dropout's best selective accuracy at 90% coverage (0.583, on a binary task) is barely better than chance after abstaining on the least-confident decile. We conclude that uncertainty-method ranking is itself shift-magnitude-dependent, not a fixed property of the method, and that no single fixed uncertainty-quantification choice can be assumed safe across the range of shift severity a deployed screening tool will encounter.

Keywords: Uncertainty quantification · Calibration · Domain shift · Selective prediction · Cervical cytology

1 Introduction

A model that fails silently – misclassifying confidently under conditions its training data did not cover – is more dangerous in a screening context than one that

fails and flags the failure. We previously established that this is not a hypothetical concern for a lightweight cervical cytology classifier: EfficientViT-M0, the best in-distribution model across a 12-model registry, is simultaneously the worst-performing model under cross-dataset domain shift, dropping below the chance floor on the more severe of two external cytology datasets tested. This paper asks the natural next question: does any standard uncertainty-quantification method's own native confidence signal correctly detect that specific failure, using only the cheapest version of each method (a single native signal, not a method-specific post-hoc correction layered on top) so the comparison reflects what a deployed system would actually have access to without extra engineering?

Our contributions are fourfold. First, a four-method uncertainty- quantification comparison (temperature-scaled softmax, MC Dropout, evidential deep learning, deep ensembles) on a backbone with a documented, severe, real (not synthetic) distribution-shift failure mode, rather than a synthetic corruption benchmark. Second, evidence that uncertainty-method ranking itself depends on shift severity: the method that wins under moderate shift (deep ensemble) is not the method that wins under severe shift (MC Dropout), and the gap between them under severe shift is large (0.473 vs. 0.285 risk-coverage AUC) while their gap under moderate shift favors the ensemble even more strongly (0.147 vs. 0.203). Third, a direct, quantified answer to whether any of the four methods reaches a *reliable* abstention policy under severe shift: none does, and we report the closest result (MC Dropout, 0.583 selective accuracy at 90% coverage) as still practically insufficient for an "abstain and refer" deployment policy, rather than rounding it up to a success. Fourth, a methods-level finding independent of the uncertainty comparison itself: MC Dropout is completely degenerate on this exact backbone unless a code-level fix is applied before training, not just at inference (Section 3.3) – a reproducibility hazard for anyone applying MC Dropout to a similarly-configured timm model without checking.

2 Related Work

Temperature scaling [1] remains the standard post-hoc softmax calibration baseline. MC Dropout [2] approximates Bayesian inference via dropout retained at test time; deep ensembles [3] instead use disagreement across independently-trained models; evidential deep learning [4] reframes classification as Dirichlet evidence accumulation, producing a single-pass uncertainty signal (vacuity) without the multi-pass or multi-model cost the other two methods incur. Adjacent work applies conformal prediction and Monte Carlo inference specifically to cervical cancer screening uncertainty; this paper differs by comparing four methods head-to-head on a documented real (not assumed) shift-failure case, rather than proposing one method in isolation.

3 Methods

3.1 Backbone

EfficientViT-M0, held fixed across all four methods to isolate the uncertainty-quantification method rather than architecture – the same backbone shown to be the best in-distribution and worst under-shift model in a 12-model registry, making it the model for which this question is most practically urgent.

3.2 Four methods, each method's own cheapest native signal

1. **Baseline**: cross-entropy training, post-hoc temperature scaling fit on a held-out calibration split; confidence = temperature- scaled max-softmax probability.
2. **MC Dropout**: dropout rate 0.3, 20 stochastic forward passes at inference; confidence = negative predictive entropy of the mean prediction across passes.
3. **Evidential deep learning**: Dirichlet evidence output via a softplus activation and the standard evidential loss (Type II MLE plus an annealed KL-divergence-to-uniform penalty); confidence = $1 -$ vacuity.
4. **Deep ensemble**: 5 independently-seeded members, each trained with the same recipe as the baseline; confidence = negative variance of the ensemble's mean predicted probability across members.

All four share one manifest, one calibration split, and one backbone initialization recipe; only the loss function and inference procedure differ.

3.3 A methods-level fix required before any GPU time was spent

The backbone's only `nn.Dropout` module ships at $p = 0$ by default in its pretrained configuration – verified directly via model inspection, not assumed – which would make MC Dropout completely degenerate (zero stochasticity across passes). A dropout rate is forced to a real, non-zero value before training begins, not just patched in at inference time, since MC Dropout's Bayesian-approximation justification specifically requires dropout to be active during training as well as at test time. We flag this because the failure mode is silent (MC Dropout would simply produce zero-variance, falsely confident "uncertainty" estimates) rather than an error.

3.4 Evaluation

All four methods evaluated identically on three splits: the SIPaKMeD in-distribution test set, Herlev (moderate shift), and Mendeley LBC (severe shift, harmonized to a shared binary normal/abnormal label). Metrics: expected calibration error (ECE), full-coverage accuracy, risk-coverage AUC (integral of error rate over the coverage curve – lower is better, since a well-behaved confidence signal keeps risk near zero until coverage is forced high), and selective accuracy at fixed coverage levels.

4 Results

Table 1. Four uncertainty methods, in distribution and under two external shifts.

Method	In-dist.		Herlev		Mendeley LBC		
	ECE	Acc.	RC-AUC	Acc.	RC-AUC	Acc.	Sel. acc.@90%
Baseline (temp.)	0.012	0.967	0.193	0.694	0.515	0.530	0.522
MC Dropout	0.013	0.972	0.203	0.658	**0.285**	0.579	**0.583**
Evidential	0.159	0.967	0.167	0.677	0.476	0.543	0.539
Deep ensemble	0.012	0.974	**0.147**	**0.771**	0.473	0.537	0.544

In distribution, three methods are close; evidential is the outlier. Baseline, MC Dropout, and the deep ensemble all calibrate well (ECE 0.012–0.013) with full-coverage accuracy in the 0.967–0.974 range. Evidential deep learning's raw belief masses are markedly less calibrated (ECE 0.159) at matched point accuracy (0.967) – its probabilities would need their own post-hoc correction, deliberately not applied here, to make this comparison's calibration numbers fair to it.

Under moderate shift (Herlev), the deep ensemble wins clearly. Its risk-coverage AUC (0.147) and full-coverage accuracy (0.771) both beat the other three methods by a real margin – member disagreement is doing genuine, useful work at this shift severity.

Under severe shift (Mendeley LBC) – the condition that matters most, since it is where this exact backbone is known to fail – the ranking inverts. MC Dropout's entropy is the only signal showing a substantial, unambiguous improvement over the uncalibrated baseline (risk-coverage AUC 0.515 to 0.285; full-coverage accuracy 0.530 to 0.579). Evidential (0.476) and the deep ensemble (0.473) are both only marginally better than the plain baseline – essentially within noise of each other – despite the ensemble's clear win on the milder shift one row above. Member disagreement, which correctly flags moderate shift, does not reliably track the specific failure mode behind the severe collapse.

No method reaches a reliable abstention operating point under severe shift. Even MC Dropout's best result – 0.583 selective accuracy at 90% coverage on a binary task – is barely better than chance after abstaining on the least-confident decile. We report this plainly rather than treating "best of four" as equivalent to "good enough": for deployment purposes, none of these four methods alone would currently justify an "abstain and refer" policy on a shift this severe.

5 Discussion

The central methodological finding is that uncertainty-method ranking is itself a function of shift severity, not a fixed property of the method. A reader who evaluated these four methods only in distribution, or only under the milder of the two shifts tested, would conclude the deep ensemble is the best choice and MC Dropout is mediocre – the opposite of what the severe-shift evaluation shows. This has a direct practical consequence: a system designer who validates an uncertainty method under one shift condition and deploys it expecting the same ranking to hold under a more severe, unseen shift is making an assumption this data does not support. It also sharpens our companion architectural finding: no architectural property predicts shift robustness, and no uncertainty-quantification property predicts shift-detection quality either, at least not one held fixed across shift severities. Together, these argue that a trustworthy deployment cannot rely on a single fixed choice at either level (backbone or uncertainty method) validated once and assumed stable.

6 Limitations

Single backbone (EfficientViT-M0); whether this specific inversion (ensemble best under moderate shift, MC Dropout best under severe shift) holds for other architectures is untested. Two shift conditions (Herlev, Mendeley LBC); "moderate" and "severe" are descriptive labels based on the accuracy drop observed, not independently validated shift-magnitude measurements. Evidential deep learning's calibration numbers are reported without the post-hoc correction its raw outputs would need for a fully fair comparison to the other three methods' already-calibrated numbers. Deep ensemble results depend on the specific 5 seeds used; seed-to-seed ensemble variance was not separately measured.

7 Conclusion

Comparing four uncertainty-quantification methods' own cheapest native confidence signals on a backbone with a documented real domain-shift failure shows that method ranking inverts between moderate and severe shift: deep ensembles win under moderate shift, MC Dropout wins under severe shift, and neither method's advantage transfers to the other condition. No method reaches a reliable abstention operating point under the severe shift condition that matters most. The practical conclusion is that a single fixed uncertainty-quantification method, chosen and validated once, cannot be assumed to remain the right choice as shift severity varies – this itself needs to be tested, not assumed, for any deployed system relying on model confidence to decide when to abstain.

References

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